

Onde Wallet API 1.0

[Document History](#)

[Introduction](#)

[Authorization](#)

[Business use cases](#)

[Testing](#)

[Go live](#)

[Troubleshooting](#)

[API Business use cases](#)

[Find all wallets](#)

[Find all wallets by criteria](#)

[Check wallet balance](#)

[Top-up wallet via external source](#)

[Manual case](#)

[Automatic case](#)

[Wallet withdraw to an external source](#)

[Objects](#)

[WalletObject](#)

[WalletMoneyObject](#)

[TopUpWithdrawRequestObject](#)

[InvoiceObject](#)

[HTTP Methods](#)

[Get all wallets of the merchant](#)

[Get specific wallet of the merchant](#)

[Request wallet top-up invoice](#)

[Request wallet withdraw invoice](#)

[Commit wallet invoice](#)

Document History

Version	Date	Change log
1.0	13.06.2022	Initial version

Introduction

Welcome! In this document, you can find API methods that reflect the functionality of Onde Wallet. Onde Wallet API allows you to integrate new payment methods for Customer Wallet top-ups on Onde platform.

Example of payment methods to be integrated:

- Mobile money
- Mobile bank application
- Web bank application
- Cryptocurrency exchange

Wallet top-ups via Onde Customer Wallet API is fully initiated by customer in 3rd party application and payment system.

Authorization

Authorization schema is similar to Onde Company & Operator APIs:

1. Authorization allows the third-party software to act on behalf of the specific Merchant Manager.
2. Authorization is a key-value pair: each request should contain a header with key name "Authorization" and value `token`.
3. Auth token is a UUID string (like driver id, company id, etc).
4. Auth token's lifetime is unlimited, until it will be replaced with a newly generated token.
5. Auth token is generated for each Merchant Manager separately in Brand user settings.

Business use cases

You can use Wallet API for these most popular cases:

1. Find all wallets
2. Find wallets by filter
3. Wallet balance check
4. Wallet top up via external source
5. Wallet withdraw

Testing

To start testing API you need to:

1. Request an invite to our sandbox environment through your Onde contact person
2. Register your company at <https://my-sandbox.onde.app>
3. Go to Brands → Merchant → User management, add a new merchant user and generate an API token
4. Pass the authorization header with your Wallet API token in every query

Notes:

- Please read the documentation carefully
- SMS verification code at Sandbox is always 000000
- Always check the Wallet API token you've generated
- {HOSTNAME} in request URLs stands for the server address. For sandbox please use `api-sandbox.onde.app`, when switching to production - `api.onde.app`
- Utilize request URLs only from the original documentation
- Header User-Agent must be present and filled in HTTP requests
- GET requests should not contain body and Content-Length Header

Go live

As soon as you finish your Wallet API integration apply for a review. You will be provided with the production endpoint and credentials afterward.

Troubleshooting

If you are facing troubles with Wallet API, please do this:

1. Ensure first that you compose requests correctly

2. Check error response messages and study how to fix requests according to this documentation

If nothing helps, please contact support@onde.app and provide as many details as your problem as you can:

1. Exact timestamp
2. URL including protocol, hostname, port, and all parameters
3. HTTP method
4. Request payload (body) for all non-GET HTTP methods
5. Request headers: used API token (last part of it)
6. Timestamp when the response was received
7. Response HTTP code
8. Response body
9. Response headers

API Business use cases

You can use Wallet API for these most popular cases:

1. Find all wallets
2. Find all wallets by criteria
3. Check wallet balance
4. Topup wallet via external source (manual case)
5. Topup wallet via external source (automatic case)
6. Withdraw wallet balance

Find all wallets

Description:

As a merchant manager, I want to see which customer & driver wallets my company has so I can prepare analytic reports, update my own customer

database, or make marketing campaigns.

Steps:

1. Generate merchant auth token for wallet API
2. Extract needed `merchantId` from MyHub URL
3. Call GET `/v1/merchant/{MERCHANT_ID}/wallet`

Find all wallets by criteria

Description:

As a merchant manager, I want to see filtered customer & driver wallets my company has so I can prepare analytic reports, find debtors or make point marketing campaigns

Steps:

1. Generate merchant auth token for wallet API
2. Extract needed `merchantId` from MyHub URL
3. Call GET `v1/merchant/{MERCHANT_ID}/wallet` with needed parameter (`phone` , `profileType` , `currency` , `gt` , `lt`)

Check wallet balance

Description:

1. As a merchant manager, I want to see the profile's wallet balance so I can see its relevant state so I can make a management decision.
2. As a third-party app/system I need to see the profile's relevant wallet balance so I can sync wallet update transactions.

Steps:

1. Generate merchant auth token for wallet API
2. Extract needed `merchantId` from MyHub URL
3. Call GET `/v1/merchant/{MERCHANT_ID}/wallet` with parameters to get needed `walletId` or extract it from your database
4. Call GET `/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}`

Top-up wallet via external source

Manual case

Description:

As a merchant manager, I want to manually top up the customer's wallet, so I can correct its balance after receiving money from the customer.

Steps:

1. generate merchant auth token for wallet API
2. extract needed `merchantId` from MyHub URL
3. After receiving money from the customer collect transaction data (phone number, profile type, currency & amount)
4. Call GET `/v1/merchant/{MERCHANT_ID}/wallet` with parameters to get needed `walletId` of your customer or extract it from your database
5. Call `/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}/topup` , mention needed amount & currency in body parameter. Remember, such method returns `topup` `invoiceId` , which needed to be committed then
6. After receiving an invoice call POST `/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}/commit` to commit the invoice

Automatic case

Description:

As a third-party app/system I need to top up the profile's wallet balance so I can sync wallet, update transactions & correct customer's balance after receiving money from the 3-rd party system.

Steps:

1. Generate merchant auth token for wallet API
2. Extract needed `merchant_Id` from MyHub URL
3. After receiving money from the 3rd-party source collect customer's data from such transaction (**separate integration needed here**)
4. Call GET `v1/merchant/{MERCHANT_ID}/wallet` with parameters to get needed `wallet_Id` of this customer (or extract it from your database)

5. Call `/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}/topup` , mention needed amount & currency in body parameter. Remember, such method returns top-up `invoiceId` , which needed to be committed then
6. After receiving an invoice, call `/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}/commit` to commit the invoice

Wallet withdraw to an external source

Description & steps for such use case are almost the same as for Top-up wallet via external source (automatic case), except money are requested to be deducted from customer's balance to 3-rd party source. So you need to use

`/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}/withdraw` method instead of `/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}/topup`

Note:

1. Withdrawal operation does not move real money from wallet, it rather decreases the balance. The real withdrawal should be carried by 3rd party integration with external payment method.
2. This method can also be used to **undo** incorrect top up operations.

Objects

WalletObject

Fields

Field name	Type	Description
walletId	string	Wallet UUID
phone	string	Profile phone number
profileType	ENUM	Customer

Field name	Type	Description
balance	List of WalletMoneyObject	List of wallet's money

Code sample

```
{
  "walletId":"12345678-1234-1234-1234-1234567890AB",
  "phone":"+123121231212",
  "profileType":"CUSTOMER",
  "balance":[
    {
      "amount":2000, // 20usd
      "currency":"USD"
    },
    {
      "amount":1000, //10eur
      "currency":"EUR"
    }
  ]
}
```

WalletMoneyObject

Fields

Field name	Type	Description
amount	Integer number	Balance money amount. Can be positive or negative. - Represents amount / 100 in decimal form (e.g., 1234 = 12.34 units) - Fractions (when expressed as decimals) must resolve to multiples of 0.01 (e.g., 12.34 is valid, 12.345 is invalid). - For whole numbers, ensure amount is a multiple of 100 (e.g., 100 = 1.00 unit, 500 = 5.00 units).

Field name	Type	Description
		Example: - To send \$15.25, use <code>amount: 1525</code> - To send €7.00, use <code>amount: 700</code> Note: Decimal values are deprecated. Use integer representation only.
Currency	ENUM	Currency code according to ISO 4217 standard.

Code sample

```
{
  "amount":2000, // 20usd
  "currency":"USD"
}
```

TopUpWithdrawRequestObject

Fields

Field name	Type	Description
money	WalletMoneyObject	See details in WalletMoneyObject
comment	string	Transaction purpose description

Code sample

```
{
  "money": {
    "amount": 2000, //20usd
    "currency": "USD"
  },
  "comment": "Bank wire transfer #1234567890.Some comment for the top u
}
```

InvoiceObject

Field name	Type	Description
invoiceId	string	Invoice UUID

Code sample

```
{
  "invoiceId": "12345678-1234-1234-1234-1234567890AB"
}
```

HTTP Methods

Get all wallets of the merchant

DESCRIPTION

This method is used to get all active customer wallets for a needed merchant, filtered by the provided parameter. To use parameters, put them into the request URL

REQUEST

1. URL: `https://{HOSTNAME}/v1/merchant/{MERCHANT_ID}/wallet`
2. Method: GET
3. Content-type: JSON

Parameters

Aa Parameter	Description	Example
<u>phone</u>	Find all wallets for selected phone number	phone=+123121233212
<u>profileType</u>	Find all wallets of customer profile type	profileType=CUSTOMER
<u>currency</u>	Find wallets, which contains needed currency	currency=[EUR,USD]
<u>gt</u>	Find all wallets which contains balances, greater than mentioned amount	gt=-5.50
<u>lt</u>	Find all wallets which contains balances, less than mentioned amount	lt=0

RESPONSE

1. Content-type: JSON
2. Body: array of [WalletObject](#)

HTTP RESPONSE CODES

Aa Code	≡ Message	≡ Solution
<u>200</u>	Standard response for successful HTTP requests.	No solution required
<u>204</u>	No content. Standard response for successful HTTP requests.	No solution required
<u>400</u>	Bad request	Check the correctness of the request fields
<u>401</u>	Authorization failure	Check your Merchant API token
<u>404</u>	Not found	Check the request URL
<u>500</u>	Internal server error	Check the correctness of the request fields
<u>502</u>	Bad gateway	Check the correctness of the request fields

RESPONSE SAMPLE

```
[
  {
    "walletId":"12345678-1234-1234-1234-1234567890AB",
    "phone":"+123121231212",
    "profileType":"CUSTOMER",
    "balance":[
      {
        "amount":2000, // 20usd
        "currency":"USD"
      },
      {
```

```

    "amount":1000, // 10eur
    "currency":"EUR"
  }
],
{
  "walletId":"12345678-1234-1234-1234-1234567890AB",
  "phone":"+123121231212",
  "profileType":"CUSTOMER",
  "balance":[
    {
      "amount":2000, // 20usd
      "currency":"USD"
    },
    {
      "amount":-5000, //-50eur
      "currency":"EUR"
    }
  ]
}
]

```

Get specific wallet of the merchant

DESCRIPTION

This method is used to get a specific active customer wallet for a needed merchant. Knowing its Id you can check wallet balance or change it.

REQUEST

1. URL: `https://{HOSTNAME}/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}`
2. Method: GET
3. Content-type: JSON

RESPONSE

1. Content-type: JSON
2. Body: `WalletObject`

HTTP RESPONSE CODES

Aa Code	≡ Message	≡ Solution
<u>200</u>	Standard response for successful HTTP requests.	No solution required
<u>204</u>	No content. Standard response for successful HTTP requests.	No solution required
<u>400</u>	Bad request	Check the correctness of the request fields
<u>401</u>	Authorization failure	Check your Merchant API token
<u>404</u>	Merchant not found / Wallet not found	Check the request URL
<u>500</u>	Internal server error	Check the correctness of the request fields
<u>502</u>	Bad gateway	Check the correctness of the request fields

RESPONSE SAMPLE

```
{
  "walletId":"12345678-1234-1234-1234-1234567890AB",
  "phone":"+123121231212",
  "profileType":"CUSTOMER",
  "balance":[
    {
      "amount":2000, //20usd
      "currency":"USD"
    },
    {
      "amount":-5000, // -50eur
      "currency":"EUR"
    }
  ]
}
```

Request wallet top-up invoice

DESCRIPTION

This method is used to increase wallet balance. Balance changing transaction consists of two stages: invoice generation & invoice commit. Such schema is implemented for fraud & doubling prevention.

REQUEST

1. URL: `https://{HOSTNAME}/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}/topup`
2. Method: POST
3. Content-type: JSON
4. Body: `TopUpWithdrawRequestObject`

RESPONSE

1. Content-type: JSON
2. Body: `InvoiceObject`

HTTP RESPONSE CODES

Aa Code	≡ Message	≡ Solution
<u>200</u>	Standard response for successful HTTP requests.	No solution required
<u>204</u>	No content. Standard response for successful HTTP requests.	No solution required
<u>400</u>	Bad request	Check the correctness of the request fields
<u>401</u>	Authorization failure	Check your Merchant API token
<u>404</u>	Merchant not found / Wallet not found	Check the request URL
<u>500</u>	Internal server error	Check the correctness of the request fields
<u>502</u>	Bad gateway	Check the correctness of the request fields

REQUEST SAMPLE

```
{
  "amount": 2000, // 20usd
  "currency": "USD"
  "comment": "Bank wire transfer #1234567890. Some comment for the top u
}
```

RESPONSE SAMPLE

```
{
  "invoiceId": "12345678-1234-1234-1234-1234567890AB"
}
```

Request wallet withdraw invoice

DESCRIPTION

This method is used to decrease wallet balance. Balance changing transaction consists of two stages: invoice generation & invoice commit. Such schema is implemented for fraud & doubling prevention.

REQUEST

1. URL: `https://{HOSTNAME}/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}/withdraw`
2. Method: POST
3. Content-type: JSON
4. Body: `TopUpWithdrawRequestObject`

RESPONSE

1. Content-type: JSON
2. Body: `InvoiceObject`

HTTP RESPONSE CODES

Aa Code	≡ Message	≡ Solution
<u>200</u>	Standard response for successful HTTP requests.	No solution required

Aa Code	≡ Message	≡ Solution
<u>204</u>	No content. Standard response for successful HTTP requests.	No solution required
<u>400</u>	Bad request	Check the correctness of the request fields
<u>400</u>	Currency=*** was not used in wallet=***** before, please select another one	There is no such currency in wallet. Check the correctness of the requested currency
<u>401</u>	Authorization failure	Check your Merchant API token
<u>404</u>	Merchant not found / Wallet not found	Check the request URL
<u>500</u>	Internal server error	Check the correctness of the request fields
<u>502</u>	Bad gateway	Check the correctness of the request fields

REQUEST SAMPLE

```
{
  "money": {
    "amount": 2000, // 20usd
    "currency": "USD"
  },
  "comment": "Charge operation #1234567890. Some comment for the withdrawal"
}
```

RESPONSE SAMPLE

```
{
  "invoiceId": "12345678-1234-1234-1234-1234567890AB"
}
```

Commit wallet invoice

DESCRIPTION

This method is used to commit the invoice of balance changing.

REQUEST

1. URL: `https://{HOSTNAME}/v1/merchant/{MERCHANT_ID}/wallet/{WALLET_ID}/commit`
2. Method: POST
3. Content-type: JSON
4. Body: `InvoiceObject`

RESPONSE

1. Content-type: JSON
2. Body: none

HTTP RESPONSE CODES

Aa Code	≡ Message	≡ Solution
<u>200</u>	Standard response for successful HTTP requests.	No solution required
<u>204</u>	No content. Standard response for successful HTTP requests.	No solution required
<u>400</u>	Bad request	Check the correctness of the request fields
<u>401</u>	Authorization failure	Check your Merchant API token
<u>404</u>	Merchant not found / Wallet not found/ Invoice not found	Check the request URL
<u>500</u>	Internal server error	Check the correctness of the request fields
<u>502</u>	Bad gateway	Check the correctness of the request fields

REQUEST SAMPLE

```
{  
  "invoiceId": "12345678-1234-1234-1234-1234567890AB"  
}
```

RESPONSE SAMPLE

1